# MSG No. 130.427 $P4'/ncc'$ [ Type III, tetragonal ]

Table 1: Wyckoff site: `4a`, site symmetry: `2.2'2'`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$ | `[1,4,11,12]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$ | `[2,3,9,10]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$ | `[5,8,15,16]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$ | `[6,7,13,14]` |

Table 2: Wyckoff site: `4b`, site symmetry: `-4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, 0\right]$ | `[1,4,13,14]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$ | `[2,3,15,16]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, 0\right]$ | `[5,8,9,10]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$ | `[6,7,11,12]` |

Table 3: Wyckoff site: `4c`, site symmetry: `4'..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4}, \frac{1}{4}, z\right]$ | `[1,4,9,10]` |
| 2 | $\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$ | `[2,3,11,12]` |
| 3 | $\left[\frac{3}{4}, \frac{3}{4}, -z\right]$ | `[5,8,13,14]` |
| 4 | $\left[\frac{1}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$ | `[6,7,15,16]` |

Table 4: Wyckoff site: `8d`, site symmetry: `-1`

| No. | position | mapping |
|---|---|---|
| 1 | $[0, 0, 0]$ | `[1,5]` |
| 2 | $\left[\frac{1}{2}, 0, \frac{1}{2}\right]$ | `[2,6]` |
| 3 | $\left[0, \frac{1}{2}, \frac{1}{2}\right]$ | `[3,7]` |
| 4 | $\left[\frac{1}{2}, \frac{1}{2}, 0\right]$ | `[4,8]` |
| 5 | $\left[\frac{1}{2}, 0, 0\right]$ | `[9,13]` |
| 6 | $\left[0, \frac{1}{2}, 0\right]$ | `[10,14]` |
| 7 | $\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$ | `[11,15]` |
| 8 | $\left[0, 0, \frac{1}{2}\right]$ | `[12,16]` |

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{3}{4}, \frac{1}{4}, z\right]$ | `[1,4]` |
| 2 | $\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2} - z\right]$ | `[2,3]` |
| 3 | $\left[\frac{1}{4}, \frac{3}{4}, -z\right]$ | `[5,8]` |
| 4 | $\left[\frac{3}{4}, \frac{1}{4}, z + \frac{1}{2}\right]$ | `[6,7]` |

*continued ...*

Table 5

| No. | position | mapping |
|---|---|---|
| 5 | $\left[\frac{1}{4}, \frac{3}{4}, z\right]$ | [9,10] |
| 6 | $\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2} - z\right]$ | [11,12] |
| 7 | $\left[\frac{3}{4}, \frac{1}{4}, -z\right]$ | [13,14] |
| 8 | $\left[\frac{1}{4}, \frac{3}{4}, z + \frac{1}{2}\right]$ | [15,16] |

Table 6: Wyckoff site: 8f, site symmetry: ..2'

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x, -x, \frac{1}{4}\right]$ | [1,12] |
| 2 | $\left[x + \frac{1}{2}, x, \frac{1}{4}\right]$ | [2,9] |
| 3 | $\left[-x, \frac{1}{2} - x, \frac{1}{4}\right]$ | [3,10] |
| 4 | $\left[\frac{1}{2} - x, x + \frac{1}{2}, \frac{1}{4}\right]$ | [4,11] |
| 5 | $\left[-x, x, \frac{3}{4}\right]$ | [5,16] |
| 6 | $\left[\frac{1}{2} - x, -x, \frac{3}{4}\right]$ | [6,13] |
| 7 | $\left[x, x + \frac{1}{2}, \frac{3}{4}\right]$ | [7,14] |
| 8 | $\left[x + \frac{1}{2}, \frac{1}{2} - x, \frac{3}{4}\right]$ | [8,15] |

Table 7: Wyckoff site: 16g, site symmetry: 1

| No. | position | mapping |
|---|---|---|
| 1 | $[x, y, z]$ | [1] |
| 2 | $\left[x + \frac{1}{2}, -y, \frac{1}{2} - z\right]$ | [2] |
| 3 | $\left[-x, y + \frac{1}{2}, \frac{1}{2} - z\right]$ | [3] |
| 4 | $\left[\frac{1}{2} - x, \frac{1}{2} - y, z\right]$ | [4] |
| 5 | $[-x, -y, -z]$ | [5] |
| 6 | $\left[\frac{1}{2} - x, y, z + \frac{1}{2}\right]$ | [6] |
| 7 | $\left[x, \frac{1}{2} - y, z + \frac{1}{2}\right]$ | [7] |
| 8 | $\left[x + \frac{1}{2}, y + \frac{1}{2}, -z\right]$ | [8] |
| 9 | $\left[\frac{1}{2} - y, x, z\right]$ | [9] |
| 10 | $\left[y, \frac{1}{2} - x, z\right]$ | [10] |
| 11 | $\left[y + \frac{1}{2}, x + \frac{1}{2}, \frac{1}{2} - z\right]$ | [11] |
| 12 | $\left[-y, -x, \frac{1}{2} - z\right]$ | [12] |
| 13 | $\left[y + \frac{1}{2}, -x, -z\right]$ | [13] |
| 14 | $\left[-y, x + \frac{1}{2}, -z\right]$ | [14] |
| 15 | $\left[\frac{1}{2} - y, \frac{1}{2} - x, z + \frac{1}{2}\right]$ | [15] |
| 16 | $\left[y, x, z + \frac{1}{2}\right]$ | [16] |